

UNIVERSITY OF
WESTMINSTER[®]



Course: Software Engineering Level 4

Module: COSC006C /Software Development 1

Type of Assignment: Individual Course Work

Submission Date: 24.12.2024

	Student ID	Student Name
IIT	20232686	Ishan Eranga Adithya Udawatte
UoW	2120028	Ishan Eranga Adithya Udawatte

Table of Contents

List of figures.....	2
Pseudocode	3
Code	9
Test Cases	19
Test Results.....	20
References.....	26

List of figures

Figure 1 Input Validation.....	20
Figure 2 Exiting	20
Figure 3 15/6/2021 Outcomes.....	21
Figure 4 15/6/2024 Histogram	21
Figure 5 16/6/2024 Outcomes.....	22
Figure 6 16/6/2024 Histogram	23
Figure 7 21/6/2024 Outcomes.....	24
Figure 8 21/6/2024 Histogram	24
Figure 9 15/6/2024.....	25
Figure 10 16/6/2024.....	25
Figure 11 21/6/2024.....	26

Pseudocode

Task D: Histogram Display Class

CLASS HistogramApp

 # Constants

 CANVAS_WIDTH = 1000

 CANVAS_HEIGHT = 600

 ATTRIBUTES constructor(traffic_data, survey_date)

 SET self.traffic_data TO traffic_data

 SET self.survey_date TO survey_date

 INITIALIZE main window using tkinter

 END FUNCTION

METHODS

 FUNCTION setup_window

 SET window size TO 1000x600

 SET window title TO "Histogram"

 CREATE canvas with CANVAS_WIDTH, CANVAS_HEIGHT and lavender background

 DISPLAY canvas

 END FUNCTION

 FUNCTION add_legend

 SET heading_text TO "Histogram of Vehicle Frequency per Hour (survey_date)"

 DISPLAY heading_text at center top of canvas

 SET legend_items TO [

 ("mediumpurple", "Elm Avenue/Rabbit Road Junction"),

 ("skyblue", "Hanley Highway/Westway Junction")

]

FOR each color, text IN legend_items

 DRAW rectangle with color

 DISPLAY text next to rectangle

END FOR

 DISPLAY "Hours 00:00 to 24:00" label on x-axis

END FUNCTION

FUNCTION draw_histogram

 SET margin TO 50

 SET bar_width TO 10

 SET bar_spacing TO 0.8

 SET category_spacing TO (CANVAS_WIDTH - 3 * margin) / 24

 SET max_height TO CANVAS_HEIGHT - 6 * margin

 # Calculate maximum value for scaling

 SET max_value TO maximum value from both junctions

 IF max_value equals 0 THEN

 DISPLAY "Warning: No traffic data available"

 SET max_value TO 1

 ENDIF

 FOR hour FROM 0 TO 23

 SET label TO formatted hour string

 GET elm_value FROM traffic_data for current hour

 GET hanley_value FROM traffic_data for current hour

 CALCULATE bar positions and heights

 # Draw bars for Elm Avenue

 DRAW rectangle for Elm Avenue data

 DISPLAY value above bar

```

        # Draw bars for Hanley Highway
        DRAW rectangle for Hanley Highway data
        DISPLAY value above bar

        # Draw hour label
        DISPLAY hour label below bars
    END FOR

    DRAW x-axis line
END FUNCTION

FUNCTION run
    CALL setup_window
    CALL add_legend
    CALL draw_histogram
    START main application loop
END FUNCTION
END CLASS

```

Task E: Multi CSV Processor Class

```

CLASS MultiCSVProcessor
    ATTRIBUTES constructor
        INITIALIZE hour_counts array of size 24 with zeros
        INITIALIZE hour_counts2 array of size 24 with zeros
    END FUNCTION

    METHODS
        FUNCTION load_csv_file(file_name)
            RESET hour_counts arrays to zero

            TRY
                OPEN file_name for reading
            
```

```

FOR each row IN file
    GET junction_name FROM row
    GET time_str FROM row

    IF time_str is not empty THEN
        CONVERT time_str to hour
        IF junction is Elm Avenue THEN
            INCREMENT hour_counts[hour]
        ELSE IF junction is Hanley Highway THEN
            INCREMENT hour_counts2[hour]
        ENDIF
    ENDIF
END FOR
RETURN true
CATCH FileNotFoundError
    RETURN false
END TRY
END FUNCTION

```

```

FUNCTION handle_user_interaction
    # Validate day input
    REPEAT
        PROMPT "Enter day (DD): "
        TRY
            READ day
            VALIDATE day between 1 and 31
            BREAK if valid
        CATCH ValueError
            DISPLAY "Integer required!"
        END TRY
    UNTIL valid day

```

```

# Validate month input
REPEAT
    PROMPT "Enter month (MM): "
    TRY
        READ month
        VALIDATE month between 1 and 12
        BREAK if valid
    CATCH ValueError
        DISPLAY "Integer required!"
    END TRY
UNTIL valid month

# Validate year input
REPEAT
    PROMPT "Enter year (YYYY): "
    TRY
        READ year
        VALIDATE year between 2000 and 2024
        VALIDATE date(year, month, day)
        BREAK if valid
    CATCH ValueError
        DISPLAY appropriate error message
    END TRY
UNTIL valid year

RETURN day, month, year
END FUNCTION

FUNCTION clear_previous_data
    RESET hour_counts arrays to zero
    CLEAR results.txt file
    DISPLAY "Previous data cleared"

```

END FUNCTION

FUNCTION process_file_name(day, month, year)

 RETURN formatted filename string

END FUNCTION

FUNCTION process_files

 DISPLAY "Traffic Flow Data Analysis"

REPEAT

 CALL handle_user_interaction TO GET day, month, year

 SET file_name TO process_file_name(day, month, year)

 SET survey_date TO formatted date string

 IF load_csv_file(file_name) successful THEN

 PROCESS csv data

 DISPLAY outcomes

 SAVE results

 CREATE traffic_data dictionary

 CREATE AND RUN HistogramApp

 ELSE

 DISPLAY "File not found" message

 CONTINUE

 ENDIF

REPEAT

 PROMPT "Load another dataset? (Y/N): "

 READ response

 IF response is "Y" THEN

 CALL clear_previous_data

 BREAK

 ELSE IF response is "N" THEN


```
        DISPLAY "Goodbye!"
        RETURN
    ENDIF
UNTIL valid response
    UNTIL user chooses to exit
END FUNCTION
END CLASS
```

```
# Main Program
BEGIN Program
    CREATE MultiCSVProcessor instance
    CALL process_files method
END Program
```

Code

#Author:I. E. A. Udawatte

#Date:23/12/2024

#Student ID:20232686 / 2120028(UOW)

Task D: Histogram Display

```
import tkinter as tk
```

```
import csv
```

```
from datetime import datetime, date
```

```
from w2120028_ABC import process_csv_data, display_outcomes, save_results_to_file
```

```
# Constants
```

```
CANVAS_WIDTH = 1000
```

```
CANVAS_HEIGHT = 600
```

```
class HistogramApp:
```

```
    def __init__(self, traffic_data, survey_date):
```

```
        """
```

```
        Initializes the histogram application with the traffic data and selected date.
```

```
        """
```

```
        self.main = tk.Tk()
```

```
        self.traffic_data = traffic_data
```

```
        self.survey_date = survey_date # Store the date for display in the heading
```

```
    def setup_window(self):
```

```
        """
```

```
        Sets up the Tkinter window and canvas for the histogram.
```

```
        """
```

```
        self.main.geometry('1000x600')
```

```
        self.main.title("Histogram")
```

```
        self.canvas = tk.Canvas(self.main, width=CANVAS_WIDTH,  
height=CANVAS_HEIGHT, background="lavender")
```

```
        self.canvas.pack()
```

```
    def add_legend(self):
```

```
        """
```

```
        Adds a legend to the histogram to indicate which bar corresponds to which junction.
```

```
        """
```

```
        # Heading of the Histogram with the date
```

```
        heading_text = f"Histogram of Vehicle Frequency per Hour ({self.survey_date})"
```

```
self.canvas.create_text((CANVAS_WIDTH / 2, 25), text=heading_text, font=("Arial",
12, "bold"))
```

```
# Creating color legend(s)
```

```
rectangle_box_y = 50
```

```
legend_items_list = [
```

```
    ("mediumpurple", "Elm Avenue/Rabbit Road Junction"),
```

```
    ("skyblue", "Hanley Highway/Westway Junction")
```

```
]
```

```
for color, text in legend_items_list:
```

```
    # Color legend box
```

```
    self.canvas.create_rectangle((50, rectangle_box_y, 60, rectangle_box_y + 10),
fill=color)
```

```
    # Color legend text
```

```
    self.canvas.create_text((70, rectangle_box_y + 5), text=text, anchor="w",
font=("Arial", 10), fill="black")
```

```
    rectangle_box_y += 20
```

```
# X-axis label
```

```
self.canvas.create_text((CANVAS_WIDTH / 2, 530), text="Hours 00:00 to 24:00",
font=("Arial", 10), fill="midnightblue")
```

```
def draw_histogram(self):
```

```
    """
```

```
    Draws the histogram with axes, labels, and bars.
```

```
    """
```

```
margin = 50
```

```
bar_width = 10
```

```
bar_spacing = 0.8 # Space between bars of each category
```

```
category_spacing = (CANVAS_WIDTH - 3 * margin) / 24 # One category for each
hour
```

```

max_height = CANVAS_HEIGHT - 6 * margin

# Get the maximum value to scale bars proportionally
max_value = max(
    max(map(int, junction.values())) for junction in self.traffic_data.values()
)

if max_value == 0:
    print("Warning: No traffic data available.")
    max_value = 1

# Draw bars
for i, hour in enumerate(range(24)):
    label = f"{hour:02}"
    elm_value = self.traffic_data["Elm Avenue/Rabbit Road"].get(label, 0)
    hanley_value = self.traffic_data["Hanley Highway/Westway"].get(label, 0)

    x_start = margin + i * category_spacing
    bar1_x1 = x_start
    bar1_x2 = bar1_x1 + bar_width
    bar2_x1 = bar1_x2 + bar_spacing
    bar2_x2 = bar2_x1 + bar_width

    bar1_height = (elm_value / max_value) * max_height
    bar2_height = (hanley_value / max_value) * max_height
    y_base = CANVAS_HEIGHT - margin - 60

    # Draw the first bar
    self.canvas.create_rectangle(bar1_x1, y_base - bar1_height, bar1_x2, y_base,
                                fill="mediumpurple", width=0)

    self.canvas.create_text(bar1_x1 + bar_width / 2, y_base - bar1_height - 10,
                            text=str(elm_value), fill="blueviolet", font=("Arial", 7))

```

```

        # Draw the second bar

        self.canvas.create_rectangle(bar2_x1, y_base - bar2_height, bar2_x2, y_base,
fill="skyblue", width=0)

        self.canvas.create_text(bar2_x1 + bar_width / 2, y_base - bar2_height - 10,
text=str(hanley_value), fill="dodgerblue", font=("Arial", 7))


        # Add label for the hour

        self.canvas.create_text((bar1_x1 + bar2_x2) / 2, y_base + 15, text=label,
fill="midnightblue", font=("Arial", 10, "bold"))


        # Draw x-axis

        self.canvas.create_line(margin, y_base, CANVAS_WIDTH - margin, y_base, width=1)


def run(self):
    """
    Runs the Tkinter main loop to display the histogram.
    """
    self.setup_window()
    self.add_legend()
    self.draw_histogram()
    self.main.mainloop()

```

Task E: Code Loops to Handle Multiple CSV Files

```
class MultiCSVProcessor:
```

```

    def __init__(self):
        """
        Initializes the application for processing multiple CSV files.
        """

        self.hour_counts = [0] * 24

        self.hour_counts2 = [0] * 24

```

```

def load_csv_file(self, file_name):
    """
    Loads a CSV file and processes its data.
    """
    # Reset counts each time data is processed
    self.hour_counts = [0] * 24
    self.hour_counts2 = [0] * 24

    try:
        with open(file_name, "r") as file:
            csv_content = csv.DictReader(file)
            for row in csv_content:
                junction_name = row.get("JunctionName", "").lower()
                time_str = row.get("timeOfDay")
                if time_str:
                    time_obj = datetime.strptime(time_str, "%H:%M:%S")
                    hour = time_obj.hour
                    if junction_name == "elm avenue/rabbit road":
                        self.hour_counts[hour] += 1
                    elif junction_name == "hanley highway/westway":
                        self.hour_counts2[hour] += 1
            return True
    except FileNotFoundError:
        return False

def handle_user_interaction(self):
    """
    Handles user input for processing multiple files.
    """

```

```
# Validate day
while True:
    try:
        self.day = int(input("Please enter the day of the survey in the format DD: "))
        if self.day < 1 or self.day > 31:
            print("Out of range - values must be in the range 1 and 31.")
            continue
        break
    except ValueError:
        print(f"Integer required!!!")
        continue
    except Exception as e:
        print(f"An error occurred: {e}")
        continue

# Validate month
while True:
    try:
        self.month = int(input("Please enter the month of the survey in the format MM: "))
        if self.month < 1 or self.month > 12:
            print("Out of range - values must be in the range 1 to 12.")
            continue
        break
    except ValueError:
        print(f"Integer required!!!")
        continue
    except Exception as e:
        print(f"An error occurred: {e}")
        continue
```

```

# Validate year
while True:
    try:
        self.year = int(input("Please enter the year of the survey in the format YYYY: "))
        if self.year < 2000 or self.year > 2024:
            print("Out of range - values must range from 2000 and 2024")
            continue
        date(self.year, self.month, self.day) # Validate if the date is valid
        break
    except ValueError as v:
        if 'invalid literal for int()' in str(v):
            print(f"Integer required!!!")
        if 'day is out of range for month' in str(v):
            print("The day is out of range for the month!., Please re-enter the date.")
        continue
    except Exception as e:
        print(f"An error occurred: {e}")
        continue

return self.day, self.month, self.year

def clear_previous_data(self):
    """
    Clears data from the previous run to process a new dataset.
    """
    self.hour_counts = [0] * 24
    self.hour_counts2 = [0] * 24
    self.current_data = None
    with open("results.txt", "w") as file:
        file.write("")

```



```

print("Previous data cleared.")

def process_file_name(self, day, month, year):
    """Generates the filename based on the provided day, month, and year."""
    file_name=f"traffic_data{day:02}{month:02}{year}.csv"
    return file_name

def process_files(self):
    """
    Main loop for handling multiple CSV files until the user decides to quit.
    """
    print("Traffic Flow Data Analysis")

    while True:
        # Get user input for file date using imported function
        day, month, year = self.handle_user_interaction()
        file_name = self.process_file_name(day, month, year)
        survey_date = f"{day:02}/{month:02}/{year}" # Format the date for display

        if self.load_csv_file(file_name):
            outcomes = process_csv_data(file_name)
            output_text = display_outcomes(outcomes)
            save_results_to_file(output_text)
            traffic_data = {
                "Elm Avenue/Rabbit Road": {f"{hour:02}": self.hour_counts[hour] for hour in
range(24)},
                "Hanley Highway/Westway": {f"{hour:02}": self.hour_counts2[hour] for hour in
range(24)},
            }
            app = HistogramApp(traffic_data, survey_date)
            app.run()

```

```

else:
    print(f"File '{file_name}' not found!!!. Please re-enter the date.")
    continue

# Ask if the user wants to load another file
while True:
    response = input("Do you want to load another dataset? (Y/N): ").strip().upper()
    if response in {"Y", "YES"}:
        self.clear_previous_data()
        break
    elif response in {"N", "NO"}:
        print("Exiting program. Goodbye!")
        return
    else:
        print("Invalid input. Please enter Y or N.")

if __name__ == "__main__":
    processor = MultiCSVProcessor()
    processor.process_files()

```

Test Cases

Test Case ID	Description	Input	Expected Output	Actual Output	Remarks
1	Validate date input - Valid date	Day: 15, Month: 06, Year: 2023	File path: "traffic_data15062023.csv" generated	File path generated successfully	Date validation working
2	Validate date input - Invalid date	Day: 15, Month: 6, Year: 1998	Error message: "Invalid Date. Please try again."	Error displayed	Invalid date handling works
3	Load CSV file - Valid file	Existing traffic data file	Dictionary with hourly counts for both junctions	Data loaded correctly	File processing successful
4	Load CSV file - Non-existent file	Non-existing file path	Empty dictionary and error message	Empty dictionary returned	Error handling works
5	Process vehicle counts	Valid traffic data	Correct vehicle counts per junction	Matches expected counts	Counter logic works
6	Draw histogram - Data visualization	Traffic data dictionary	Histogram with colored bars	Visual representation correct	Graph rendering works
7	Validate numeric input - Valid range	Input: 5 (min:1, max:31)	Value accepted	5 accepted	Range validation works
8	Validate numeric input - Out of range	Input: 32 (min:1, max:31)	Error: "Value must be between 1 and 31"	Error displayed	Out of range handling works
9	Clear previous data	After processing file	All counters reset to zero	Data cleared	Reset functionality works
10	Legend display	Valid traffic data	Legend showing both junctions	Legend displayed correctly	Visual elements correct

Test Results

```
>>> = RESTART: C:\Eranga\IIT\Lectures\Degree 1st Year\4COSC006C [DESI] Software Development
I\Lec\week 10\w2120028\w2120028_DE.py
Traffic Flow Data Analysis
Please enter the day of the survey in the format DD: 32
Out of range - values must be in the range 1 and 31.
Please enter the day of the survey in the format DD: UGU
Integer required!!!
Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 13
Out of range - values must be in the range 1 to 12.
Please enter the month of the survey in the format MM: fccyg
Integer required!!!
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: fygfv
Integer required!!!
Please enter the year of the survey in the format YYYY: 1998
Out of range - values must range from 2000 and 2024
Please enter the year of the survey in the format YYYY: 2021
File 'traffic_data15062021.csv' not found!!!. Please re-enter the date.
Please enter the day of the survey in the format DD: |
```

Figure 1 Input Validation

```
Do you want to load another dataset? (Y/N): n
Exiting program. Goodbye!
|
```

Figure 2 Exiting

```

Please enter the day of the survey in the format DD: 15
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024

Here is the requested traffic data.

*****
The selected data file: traffic_data15062024.csv
*****

The total number of vehicles for this date: 1037
The total number of trucks for this date: 109
The total number of electric vehicles for this date: 368
The total number of "two wheeled" vehicles for this date: 401
The total number of busses leaving Elm Avenue/Rabbit Road heading north: 15
The total number of vehicles passing through both junctions without turning left or right: 363
The percentage of total vehicles recorded that are Trucks for this date: 11%
The average number Bicycles per hour for this date: 7
The total number of vehicles recorded as over the speed limit for this date: 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction: 494
The total number of vehicles recorded through Hanley Highway/Westway junction: 543
10% of vehicles recorded through Elm Avenue/Rabbit Road that are Scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway: 39
The most vehicles through Hanley Highway/Westway were recorded between: 18:00 between 19:00
The number of hours of rain for this date: 0

*****

```

Figure 3 15/6/2021 Outcomes

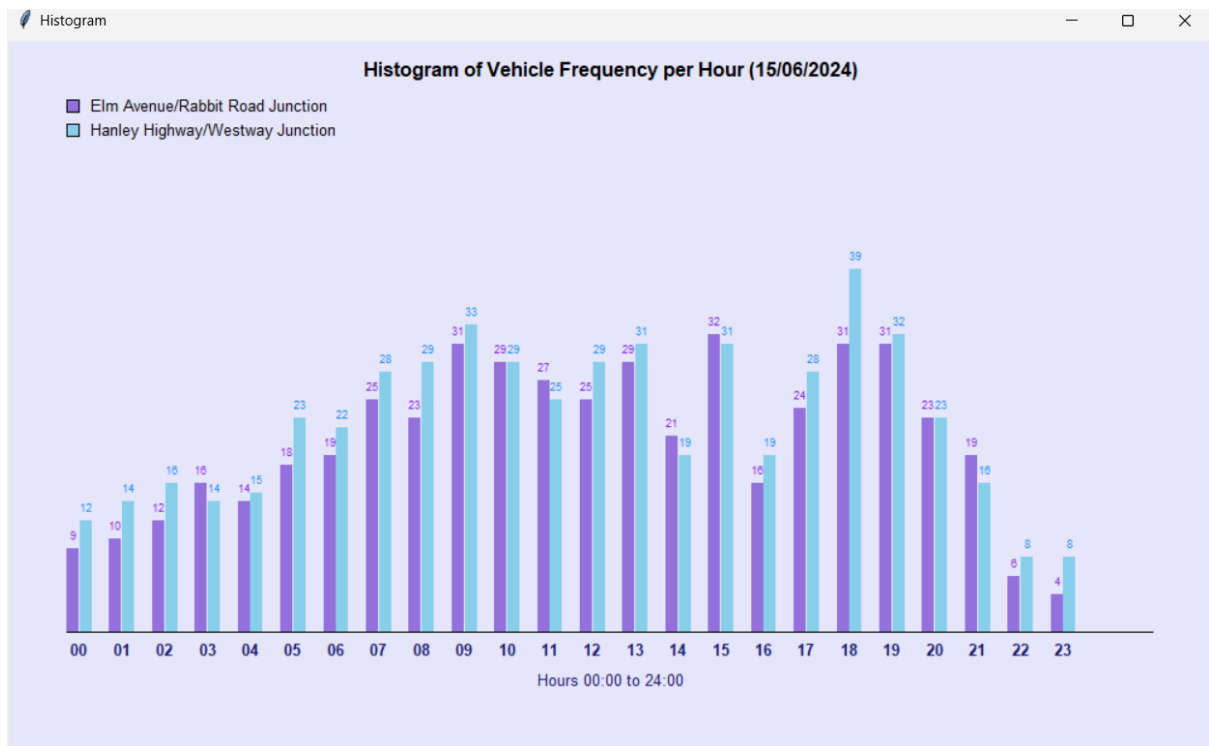


Figure 4 15/6/2024 Histogram

```
Do you want to load another dataset? (Y/N): y
Previous data cleared.
Please enter the day of the survey in the format DD: 16
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024

Here is the requested traffic data.

*****
The selected data file: traffic_data16062024.csv
*****

The total number of vehicles for this date: 101
The total number of trucks for this date: 11
The total number of electric vehicles for this date: 29
The total number of "two wheeled" vehicles for this date: 29
The total number of busses leaving Elm Avenue/Rabbit Road heading north: 0
The total number of vehicles passing through both junctions without turning left or right: 38
The percentage of total vehicles recorded that are Trucks for this date: 11%
The average number Bicycles per hour for this date: 0
The total number of vehicles recorded as over the speed limit for this date: 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction: 52
The total number of vehicles recorded through Hanley Highway/Westway junction: 49
5% of vehicles recorded through Elm Avenue/Rabbit Road that are Scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway: 5
The most vehicles through Hanley Highway/Westway were recorded between: 1:00 between 2:00
The number of hours of rain for this date: 3

*****
```

Figure 5 16/6/2024 Outcomes

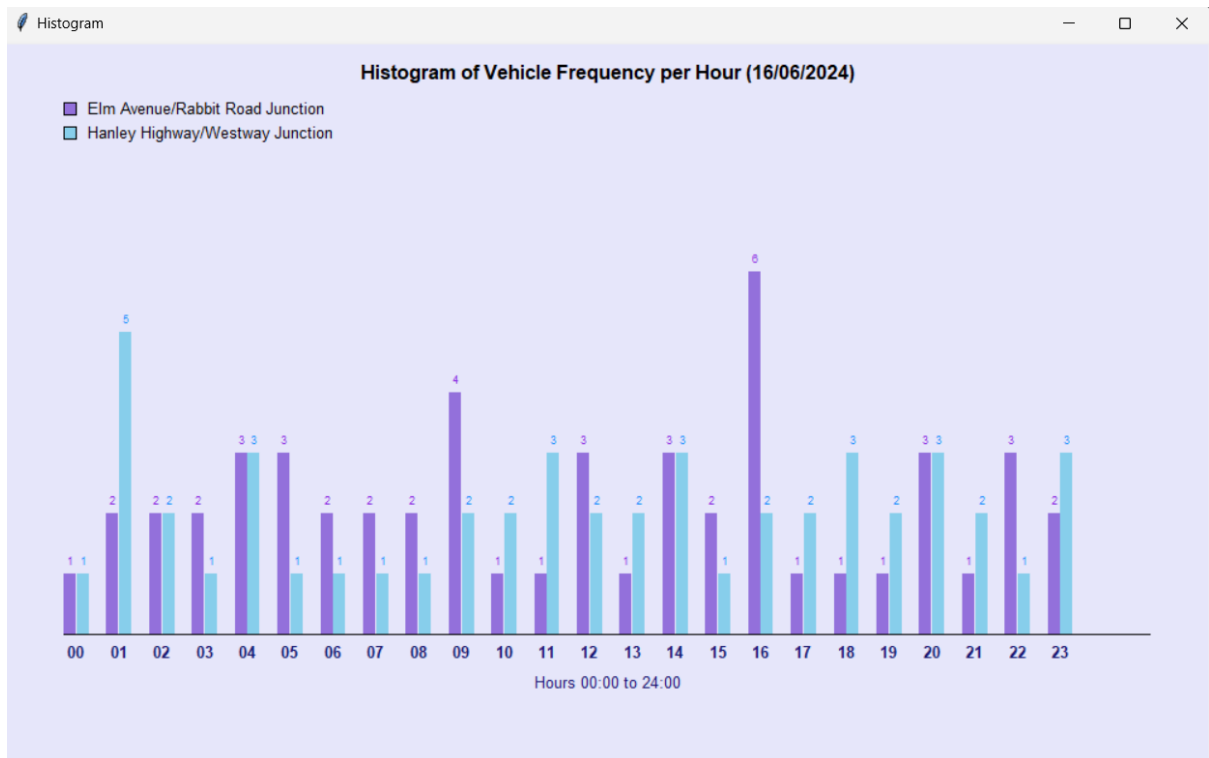


Figure 6 16/6/2024 Histogram

```

Do you want to load another dataset? (Y/N): y
Previous data cleared.
Please enter the day of the survey in the format DD: 21
Please enter the month of the survey in the format MM: 6
Please enter the year of the survey in the format YYYY: 2024

Here is the requested traffic data.

*****
The selected data file: traffic_data21062024.csv
*****

The total number of vehicles for this date: 1334
The total number of trucks for this date: 138
The total number of electric vehicles for this date: 442
The total number of "two wheeled" vehicles for this date: 503
The total number of busses leaving Elm Avenue/Rabbit Road heading north: 19
The total number of vehicles passing through both junctions without turning left or right: 494
The percentage of total vehicles recorded that are Trucks for this date: 10%
The average number Bicycles per hour for this date: 10
The total number of vehicles recorded as over the speed limit for this date: 250
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction: 651
The total number of vehicles recorded through Hanley Highway/Westway junction: 683
9% of vehicles recorded through Elm Avenue/Rabbit Road that are Scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway: 71
The most vehicles through Hanley Highway/Westway were recorded between: 18:00 between 19:00
The number of hours of rain for this date: 6

*****

```

Figure 7 21/6/2024 Outcomes

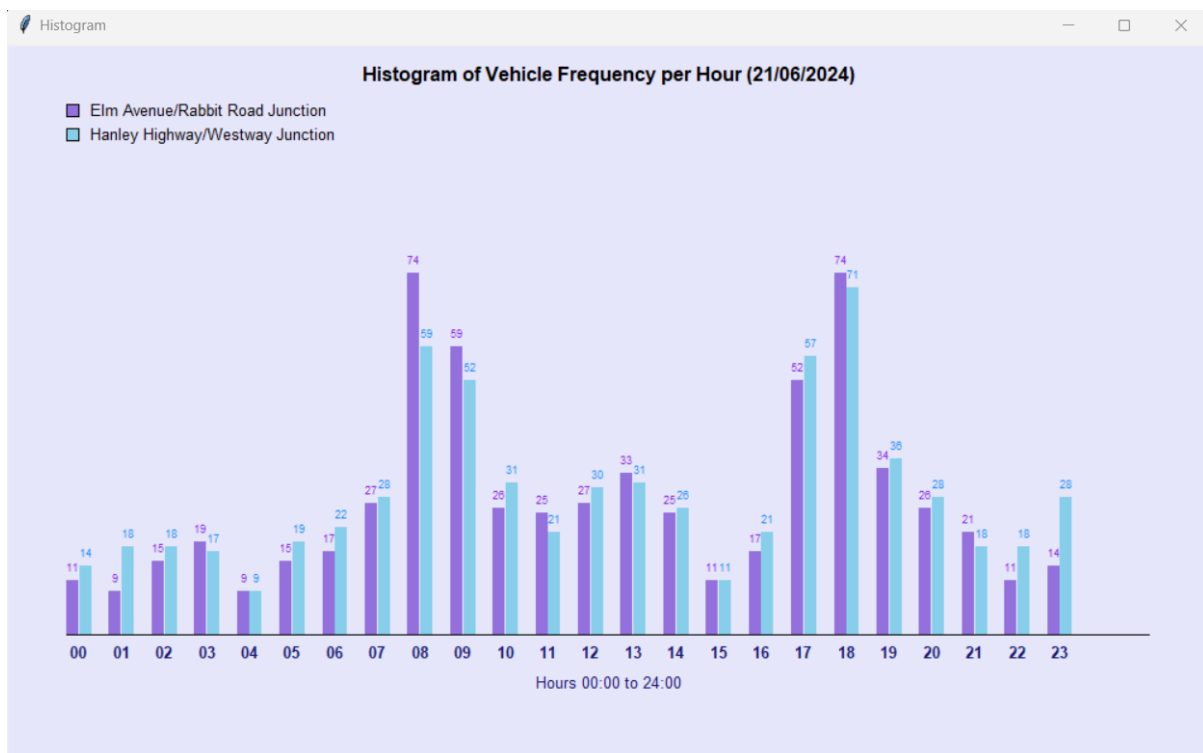


Figure 8 21/6/2024 Histogram


```
results.txt
File Edit View

Here is the requested traffic data.

*****
The selected data file: traffic_data15062024.csv
*****

The total number of vehicles for this date: 1037
The total number of trucks for this date: 109
The total number of electric vehicles for this date: 368
The total number of "two wheeled" vehicles for this date: 401
The total number of busses leaving Elm Avenue/Rabbit Road heading north: 15
The total number of vehicles passing through both junctions without turning left or right: 363
The percentage of total vehicles recorded that are Trucks for this date: 11%
The average number Bicycles per hour for this date: 7
The total number of vehicles recorded as over the speed limit for this date: 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction: 494
The total number of vehicles recorded through Hanley Highway/Westway junction: 543
10% of vehicles recorded through Elm Avenue/Rabbit Road that are Scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway: 39
The most vehicles through Hanley Highway/Westway were recorded between: 18:00 between 19:00
The number of hours of rain for this date: 0

*****
```

Figure 9 15/6/2024

```
results.txt
File Edit View

Here is the requested traffic data.

*****
The selected data file: traffic_data16062024.csv
*****

The total number of vehicles for this date: 101
The total number of trucks for this date: 11
The total number of electric vehicles for this date: 29
The total number of "two wheeled" vehicles for this date: 29
The total number of busses leaving Elm Avenue/Rabbit Road heading north: 0
The total number of vehicles passing through both junctions without turning left or right: 38
The percentage of total vehicles recorded that are Trucks for this date: 11%
The average number Bicycles per hour for this date: 0
The total number of vehicles recorded as over the speed limit for this date: 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction: 52
The total number of vehicles recorded through Hanley Highway/Westway junction: 49
5% of vehicles recorded through Elm Avenue/Rabbit Road that are Scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway: 5
The most vehicles through Hanley Highway/Westway were recorded between: 1:00 between 2:00
The number of hours of rain for this date: 3

*****

Ln 13, Col 5 1,514 characters 100% Windows (CRLF) ANSI
```

Figure 10 16/6/2024

```
Here is the requested traffic data.

*****
The selected data file: traffic_data21062024.csv
*****

The total number of vehicles for this date: 1334
The total number of trucks for this date: 138
The total number of electric vehicles for this date: 442
The total number of "two wheeled" vehicles for this date: 503
The total number of busses leaving Elm Avenue/Rabbit Road heading north: 19
The total number of vehicles passing through both junctions without turning left or right: 494
The percentage of total vehicles recorded that are Trucks for this date: 10%
The average number Bicycles per hour for this date: 10
The total number of vehicles recorded as over the speed limit for this date: 250
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction: 651
The total number of vehicles recorded through Hanley Highway/Westway junction: 683
9% of vehicles recorded through Elm Avenue/Rabbit Road that are Scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway: 71
The most vehicles through Hanley Highway/Westway were recorded between: 18:00 between 19:00
The number of hours of rain for this date: 6

*****
```

Figure 11 21/6/2024

References

<https://www.geeksforgeeks.org/>

<https://www.w3schools.com/>

<https://gemini.google.com/>